

THE REALITY ENGINE

GAME DESIGN PARAMETERS & APPROACH

The Complete Design Bible

Architecture, philosophy, reward mechanics, content structure, and implementation decisions for the world's first AI-guided infrastructure education platform.

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1. DESIGN PHILOSOPHY

The Reality Engine is not a game that teaches. It is a teaching system that uses game mechanics as delivery vehicles. The distinction matters because it determines every design decision that follows. The content — 12 Civilisational Relays, 91+ inventions, the ISI framework, the Survival Formula — is fixed, authoritative, and non-negotiable. The **wrappers** around that content are where all creative latitude exists.

The core design philosophy rests on three pillars. First, **no loss states** — the player can never fail, die, lose progress, or be punished. Every action yields a reward. The variable is magnitude, not existence. Second, **big numbers** — XP rewards are deliberately large (10,000–50,000 per action) because the system respects the player's time and effort. Small numbers feel dismissive; large numbers feel generous. The psychological effect of earning 50,000 XP for a single discovery is fundamentally different from earning 50. Third, **the same knowledge, different wrappers** — all modes, all age groups, all pathways lead to the same 12 Relays and the same 91+ inventions. The content never changes. Only the presentation layer changes.

This philosophy draws from three proven design traditions. The **variable ratio reinforcement schedule** from B.F. Skinner's operant conditioning research [1] powers the Relay Spinner — the same mechanism that makes slot machines the most engagement-resistant devices ever built, repurposed for education instead of extraction. The **room-by-room dungeon crawl** from Tom Moldvay's 1981 Basic D&D [2] powers the Dungeon Crawl — the same structure that taught millions of children to make decisions, imagine consequences, and explore the unknown. The **transformation ritual** from Filmation's He-Man [3] powers the Power of Grey Matter — the same emotional arc that gives a 12-year-old the visceral feeling of becoming something greater through effort and knowledge.

2. CONTENT ARCHITECTURE

2.1 The 12 Civilisational Relays

The relays are the atomic units of the entire system. They are chronologically ordered, historically grounded, and collectively exhaustive — covering the full arc of human infrastructure from controlled fire (pre-10,000 BCE) to artificial intelligence and collective consciousness (2025+). Each relay has a name, a subtitle, an emoji symbol, an era, a web classification, and a colour.

#	Relay	Subtitle	Era	Web	Colour
1	Fire	The Eternal Constant	pre-10,000 BCE	Natural	#ef4444
2	Tree	The Living Foundation	10,000–5,000 BCE	Natural	#22c55e
3	River	Cradles of Continuity	5,000–3,000 BCE	Natural	#3b82f6
4	Horse	The Velocity of Intent	3,000–1,000 BCE	Machine	#d97706
5	Roads	Arteries of Intent	1,000 BCE–500 CE	Machine	#a16207
6	Ships	The Master Weaver's Reach	500–1500 CE	Machine	#0891b2
7	Loom	The Binary Birth	1500–1760 CE	Digital	#9333ea
8	Rail	Standardizing the Continental Rhythm	1760–1870 CE	Machine	#64748b
9	Engine	The Internal Revolution	1870–1914 CE	Machine	#b45309
10	AAA Triad	The Triple Convergence	1914–1969 CE	Digital	#0284c7
11	Orbit	The Programmable Frontier	1969–2025 CE	Digital	#6366f1
12	Human Nodes	The Torus Metaphor	2025+ CE	Consciousness	#a855f7

2.2 The Five Great Webs

The webs are the classification system that groups relays by domain. They also serve as the alignment axis for the FITS temperament system and the MPNC Fleet craft selection.

Web	Icon	Colour	Relays	Domain
Natural		#22c55e	1–3	Biological systems, ecosystems, natural forces
Machine		#f59e0b	4–6, 8–9	Engines, structures, physical infrastructure
Digital		#3b82f6	7, 10–11	Communications, computing, signal networks
Biological		#ec4899	Cross-relay	Medicine, genetics, biological engineering
Consciousness		#a855f7	12	Awareness, philosophy, collective intelligence

The Biological Web is unique in that it does not map to a contiguous set of relays. Instead, it threads through multiple relays — herbal medicine in the Fire era, anatomical study in the River era, vaccination in the Engine era, genetics in the Orbit era. This cross-cutting nature makes it the "healer's web" — it sustains and repairs what the other webs build.

2.3 The 91+ Inventions

Each relay contains between 5 and 12 inventions — real historical discoveries, technologies, or systems that changed the trajectory of civilisation. The "91+" designation indicates a minimum count; the system is designed to accommodate additional inventions as the content library expands. Inventions are the collectible units of the game — each one has an **iCard** with its name, relay, era, description, and historical significance.

2.4 The ISI Framework

The Infrastructure Significance Index (ISI) is the mathematical backbone of the entire system. It is documented in full in TP-016 [4] and operates across three dimensions:

Dimension	Name	What It Measures
ISI ₁	Sustainability	Long-term viability of infrastructure knowledge

Dimension	Name	What It Measures
ISI ₂	Survival	The Survival Formula: $S = (A \times P) / \beta$
ISI ₃	Significance	Innate value and economic impact of infrastructure

The **Survival Formula** $S = (A \times P) / \beta$ is the central equation. **A** (Accuracy) measures whether the right things are being learned. **P** (Precision) measures how well they are being learned. **β** (the Human Blip) is the constant 0.000267% — the fraction of Earth's history that contains all of human civilisation. The formula yields the **Survival Index**, which maps to the Civilisation Clock. The higher S goes, the further the clock retreats from midnight.

The **mobilisation target** is 1 billion learners (12.1% of the global population), each contributing 2.58% of their day (2,229 seconds / 37 minutes 9 seconds). This target is not arbitrary — it is derived from the ISI₂ calculation as the threshold at which the collective counterweight exceeds the civilisational risk.

3. PLAYER PROGRESSION SYSTEM

3.1 Age-Tiered Modes

The Reality Engine uses a three-tier age progression, with the Explorer tier further subdivided into four sub-modes. This structure ensures that the same content is accessible to an 8-year-old and a 28-year-old, with the wrapper complexity scaling to match cognitive development.

Tier	Mode	Age Range	Complexity	Session Length
Explorer	Classic Explorer	8–14	Minimal	2–10 min
Explorer	Relay Spinner	8–10	Low	2–5 min
Explorer	Dungeon Crawl	10–12	Medium	10–20 min
Explorer	Power of Grey Matter	12–14	Medium-High	15–30 min
Intermediate	Flight Deck	14–18	High	20–45 min

Tier	Mode	Age Range	Complexity	Session Length
Advanced	Scholar	18+	Full Academic	30–90 min

The progression path is strictly linear: Explorer → Flight Deck → Scholar → The World. However, within each tier, players can move freely between sub-modes without losing progress. Inventions discovered in Spinner mode count in Dungeon mode. Powers earned in Grey Matter mode appear in the collection grid. The modes are **views**, not **silos**.

3.2 XP and BitPoint Economy

The XP system uses deliberately large numbers to convey respect for the player's effort. The conversion rate is fixed at 100 XP = 1 BitPoint.

Milestone	XP Required	Title
Entry	0	Apprentice
First Gate	100,000	Relay Scout
Momentum	500,000	Relay Runner
Mastery	1,000,000	GURU
Elite	5,000,000	Grand Master
Cap	24,000,000	Odyssey Complete

The 24,000,000 XP cap is a hard ceiling. It represents the theoretical maximum a player can earn by completing every discovery, every challenge, every mission across all modes. The cap exists to prevent inflation and to give the "Odyssey Complete" title genuine scarcity.

3.3 Reward Rates by Mode

Each mode awards XP at different rates, calibrated to the effort required per action:

Event	Spinner	Dungeon	Grey Matter
Single invention discovery	50,000	25,000	30,000
Partial discovery / lore fragment	10,000	10,000	10,000
Puzzle / quiz completion	—	15,000	20,000
Relay completion	100,000	150,000	200,000
Power earned	—	—	100,000
Full game completion	1,200,000	1,800,000	2,400,000
Daily streak bonus (per day)	25,000	25,000	25,000

Grey Matter awards the most XP because it demands the most effort (discovery + knowledge + application). Spinner awards the least per action but has the highest action frequency. Dungeon sits in the middle. All three are designed so that a dedicated player can reach the 1M GURU threshold within 2–3 weeks of daily play.

3.4 Daily Engagement Mechanics

The daily engagement system uses two mechanisms. **Daily Spins** in Relay Spinner mode give 12 free spins per day (one per relay). Additional spins are earned through streak bonuses (3 consecutive days = 6 bonus spins). **Daily Streaks** across all modes award 25,000 bonus XP per consecutive day of play. There is no paid spin mechanism and no paid acceleration — this is education, not extraction.

4. FITS TEMPERAMENT SYSTEM

The FITS (Feeler, Intuitive, Thinker, Sensor) system is the personality framework that aligns players to webs, craft, and civilisational perspectives. It is introduced subtly in Explorer mode (through DAVID's observation of player choices) and becomes explicit in Flight Deck mode (through craft selection and the Dearden Field).

FITS Type	Abbreviation	Description	Web	Craft
Senser	S	Scouts ahead, reads terrain, maps the unknown	Natural	TB-1 Pathfinder
Thinker	T	Builds, engineers, constructs — civilisation follows	Machine	TB-2 Forgemaster
Intuitive	I	Connects, communicates, weaves the digital web	Digital	TB-3 Starcaster
Feeler	F	Heals, sustains, carries the conscience of the fleet	Biological	TB-4 Lifeline
Balanced	B	Commands, coordinates, holds the field together	Consciousness	TB-5 Sentinel

The fifth type, **Balanced**, is not a FITS type in the traditional sense — it represents the player who scores evenly across all four dimensions. The Balanced player gravitates toward the Consciousness Web and the command role.

5. MPNC FLEET — THE THUNDERBIRD CRAFT

The MPNC (Multi-Purpose Navigation Craft) Fleet consists of five Thunderbird-class vessels, each aligned to a FITS type and a Great Web. Craft selection occurs in Flight Deck mode and determines the player's XP bonus structure, visual identity, and mission specialisation.

Craft	Class	Pilot	Role	Stats Profile
TB-1 Pathfinder	Arrow-Class	Scott / IQ	Rapid-Response Reconnaissance	Speed 9, Sensors 8, Range 10
TB-2 Forgemaster	Griffin-Class	Virgil / EQ	Heavy Engineering & Construction	Armour 9, Range 7, Sensors 6
TB-3 Starcaster	Zeta-Class	Alan / CQ	Communications & Signal Intelligence	Sensors 10, Range 9, Stealth 8
TB-4 Lifeline	Leviathan-Class	Gordon / Deep CQ	Medical & Humanitarian Response	Range 8, Sensors 7, Speed 6
TB-5 Sentinel	Citadel-Class	John / HQ	Command & Control / Overwatch	Armour 8, Sensors 9, Speed 3

Each craft provides a **+20% XP bonus** on relays aligned to its web (TB-5 Sentinel provides +15% on Consciousness Relays). The five stats — Speed, Armour, Sensors, Range, Stealth — are displayed as bar charts on the craft selection card and influence mission outcomes in Flight Deck mode.

6. ABILITY SCORE SYSTEMS

The Reality Engine uses two ability score systems, scaled to the player's age tier.

6.1 Explorer Ability Scores (Dungeon Crawl)

The simplified system uses three scores mapped to the three civilisational perspectives:

Ability	Abbreviation	Perspective	Mechanic
Observation	OBS	Western — analytical, systematic	Bonus XP on puzzle rooms
Intuition	INT	Eastern — holistic, connective	Reveals hidden rooms
Resilience	RES	Nomadic — adaptive, survival	Extra retries on failed checks

Each score is set by a single d20 roll (max 3 re-rolls for the full set). Scores range from 1–20 with simple modifiers: 1–5 = -1, 6–15 = 0, 16–20 = +1.

6.2 Scholar Ability Scores (Full System)

The complete system uses six scores for the Scholar and Flight Deck tiers:

Ability	Abbreviation	Domain
Observation	OBS	Perception and awareness of infrastructure patterns
Analysis	ANA	Logical reasoning and systems thinking
Synthesis	SYN	Combining knowledge across domains

Ability	Abbreviation	Domain
Communication	COM	Articulating ideas and persuading others
Implementation	IMP	Practical application and project execution
Vision	VIS	Strategic foresight and civilisational thinking

These six scores are rolled using the full d20 protocol and influence mission outcomes, thesis grading, and the Dearden Field navigation in Scholar mode.

7. THE DEARDEN FIELD

The Dearden Field is the 60-node matrix that forms the core navigation system of Flight Deck mode. It is constructed by intersecting the 12 Relays with the 5 Great Webs, producing a 12×5 grid of 60 nodes. Each node represents a unique intersection — for example, "Fire × Natural" (how fire shaped ecosystems) or "Loom × Digital" (how weaving patterns led to binary code).

The Dearden Field is not just a navigation tool — it is a **knowledge map**. Completing nodes reveals connections between relays and webs that are not visible in the linear relay sequence. The field is displayed as an interactive HUD in the Flight Deck cockpit, with completed nodes glowing and uncompleted nodes dimmed.

8. DAVID — THE AI COMPANION

DAVID (**D**igital **A**ugmented **V**isual **I**ntelligence **D**isplay) is the AI companion that accompanies the player across all modes. DAVID is not a chatbot — he is a **narrator, guide, dungeon master, and guardian** whose personality shifts to match the mode context.

Mode	DAVID's Role	Voice Register
Classic Explorer	Friendly narrator	Warm, encouraging

Mode	DAVID's Role	Voice Register
Relay Spinner	Post-spin commentator	Excited, celebratory
Dungeon Crawl	Dungeon Master	Atmospheric, descriptive
Power of Grey Matter	Ancient Guardian	Dramatic, mythic
Flight Deck	Co-pilot	Professional, tactical
Scholar	Academic Advisor	Formal, analytical

DAVID uses the platform's LLM integration for contextual responses and the Web Speech API for voice narration. The voice can be toggled on or off by the player. DAVID tracks player behaviour patterns — which paths they choose, which relays they gravitate toward, which ability scores they rely on — and uses this data to reveal the player's **civilisational perspective pattern** at the end of their journey.

9. VISUAL AND AUDIO DESIGN

9.1 Visual Style by Mode

Each mode has a distinct visual identity, but all share the same colour palette derived from the relay and web colours defined in the content architecture.

Mode	Visual Style	Reference
Classic Explorer	Clean, modern, tap-friendly tiles	Mobile-first discovery UI
Relay Spinner	Bright arcade-cabinet aesthetic, chunky symbols, coin-shower animations	1980s arcade meets modern mobile
Dungeon Crawl	Illustrated text adventure, parchment panels, serif typography	Moldvay Basic D&D (1981)
Power of Grey Matter	Bold, heroic, primary colours, transformation animations	He-Man / Masters of the Universe

Mode	Visual Style	Reference
Flight Deck	Dark cockpit HUD, glowing nodes, tactical overlays	Sci-fi command bridge
Scholar	Academic, structured, thesis-grade typography	University assessment system

9.2 Audio Design

The platform uses three audio layers. **DAVID's voice** provides narration and guidance via Web Speech API with a K-Pop Guide voice profile — warm, clear, and energetic. **Background music** is mode-specific: upbeat anime-style EDM for Explorer modes, atmospheric ambient for Dungeon Crawl, heroic brass for Grey Matter transformations. **Sound effects** include mechanical reel clicking (Spinner), dungeon ambience (Dungeon Crawl), energy gathering and lightning strikes (Grey Matter transformations), and cockpit beeps (Flight Deck).

9.3 Youth Intro Videos

Each Explorer sub-mode has a short (20–30 second) animated intro video embedded at the top of its page, accessible via a "Watch Intro" button. The videos use cosmic anime-style visuals with DAVID's narration and upbeat BGM. They translate the ISI/Blip concepts into age-appropriate language — time, games, superpowers, teamwork — without academic jargon.

Video	Duration	Mode	Core Message
Spin the Story of Everything	20s	Relay Spinner	"Everything you use was invented by someone"
Your Adventure Through 12,000 Years	26s	Dungeon Crawl	"Explore a cave where humans first discovered fire"
Unlock the Powers That Built Everything	30s	Power of Grey Matter	"Every invention can give you a superpower"

10. PLATFORM ARCHITECTURE

10.1 Site Map

The Reality Engine is a single-page application with the following route structure:

Route	Page	Purpose
/	Home	Landing page with mode selection and mission statement
/explore	Explorer Select	Sub-mode selection (Classic, Spinner, Dungeon, Grey Matter)
/explore/spinner	Relay Spinner	Slot machine discovery game
/explore/dungeon	Dungeon Crawl	Room-by-room RPG exploration
/explore/greymatter	Power of Grey Matter	Power-up transformation missions
/explore/:relayNum	Explorer Relay	Classic tap-to-discover for a specific relay
/m/explore	Mobile Explorer	Mobile-optimised discovery experience
/flight-deck	Flight Deck	Cockpit HUD with Dearden Field and craft selection
/create	Scholar Create	Full character creation for Scholar mode
/leaderboard	Leaderboard	Live global rankings
/cards	Card Collection	iCard gallery and collection tracker
/synthesis	Synthesis	End-game pattern reveal and journey summary
/journey	Journey	Player's personal timeline and milestones
/challenge/:code	Challenge Landing	Shareable challenge entry points
/network	Network Directory	Community and contributor directory
/media	Media Gallery	Video, image, and audio content library
/bridges	Bridge Hub	Tetrahedral Observer — connections to partner sites

Route	Page	Purpose
/frameworks	Frameworks	ISI, FITS, and other conceptual frameworks
/governance	Governance Deck	SAP-001 protocol and governance cards
/yoda	Yoda Control	Administrative control panel

10.2 Shared Backend Infrastructure

All modes share the same backend infrastructure. The database tables — [player_profiles](#), [relay_progress](#), [discoveries](#), [xp_transactions](#) — are mode-agnostic. The content source ([shared/gameData.ts](#)) provides the same 12 Relays, 91+ inventions, 5 Webs, 5 FITS types, 5 Craft, and 6 Ability Scores to all modes. The XP constants (24M cap, 100:1 BitPoint rate, GURU at 1M) are universal. DAVID's LLM backend is shared, with mode-specific system prompts determining his personality.

Progress is persistent and cross-modal. A player who discovers an invention in Spinner mode will see it marked as discovered in Dungeon mode. A player who earns a power in Grey Matter mode will see it reflected in their collection grid. The modes are **views**, not **silos**.

10.3 The Tetrahedral Observer

The Reality Engine exists within a four-site ecosystem called the **Tetrahedral Observer**:

Bridge	Name	Domain	Role
ACAD SITE	Infrastructure Academy	infra-acad-kuqzaex2.manus.space	Master database, academic content
MEMORIAL SITE	Principia Tectonica	nigelmemorial-ucmtq9dn.manus.space	Narrative archive, memorial content
TRE GAME	The Reality Engine	realityeng-epdhlkrn.manus.space	Interactive game platform
CHART ROOM	The Chartered Chart	xgrowthtrk-2a93yo5z.manus.space	Analytics and growth tracking

11. DESIGN DECISIONS LOG

This section records the key design decisions made during development, with rationale for each.

Decision	Rationale	Date
5 Webs instead of 3+1	The Biological Web needed explicit recognition as a cross-cutting domain, not a footnote. Medicine, genetics, and bioengineering deserve their own classification.	22 Mar 2026
"Grey Matter" not "Grayskull"	The He-Man reference was the design inspiration, but the actual mode name needed to be original IP. "Grey Matter" preserves the transformation mechanic while being legally clean and thematically accurate (grey matter = brain = intelligence).	22 Mar 2026
No loss states anywhere	Every spin, every room, every mission yields a reward. The variable is magnitude, not existence. This eliminates frustration while preserving the engagement loop. Children should never feel punished for trying.	22 Mar 2026
Big XP numbers	50,000 XP for a single discovery feels generous and respectful. 50 XP for the same discovery feels dismissive. The psychological impact of large numbers drives engagement without any cost to the system.	22 Mar 2026
24M XP hard cap	Prevents inflation, gives "Odyssey Complete" genuine scarcity, and provides a clear finish line for completionists.	22 Mar 2026
3 simplified ability scores for Dungeon Crawl	The full 6-score system is too complex for 10–12 year olds. OBS/INT/RES maps cleanly to the three civilisational perspectives (Western/Eastern/Nomadic) and keeps the maths trivial.	22 Mar 2026
Modes as views, not silos	Cross-modal progress persistence means players never lose work by switching modes. This encourages experimentation and reduces commitment anxiety.	22 Mar 2026
DAVID voice toggle	Not all players want audio narration. The toggle respects player preference while keeping the feature available for those who benefit from it.	22 Mar 2026
Youth intro videos per sub-mode	8–14 year olds need visual hooks, not text explanations. A 20-second video communicates more than a 500-word description.	22 Mar 2026
Craft images on selection cards	Visual identity is critical for the Flight Deck experience. Abstract icons were insufficient — players need to see their craft to feel ownership.	22 Mar 2026

12. RELATIONSHIP TO THE BOOK SERIES

The Reality Engine is the interactive companion to **An Infrastructure Odyssey** — the book series structured as Episode 1: Calories to Consciousness (Book 1 of 3). The book has three sections: **The Perspective** (narrative foundation), **The Guide** (practical framework following the 4-Pillar Framework: Observational → Educational → Application → Thesis), and **The Game** (this platform).

The 12 Relays in the game correspond directly to the 12 chapters of the book. The ISI framework is documented in the Guide section. The FITS temperament system, the Dearden Field, and the Survival Formula all originate in the book and are implemented interactively in the game. The game is not a separate product — it is Part 3 of the same work, delivered through a different medium.

REFERENCES

[1]: Skinner, B.F. (1957). *Schedules of Reinforcement*. Appleton-Century-Crofts. Variable ratio schedules produce the highest, most consistent response rates.

[2]: Moldvay, T. (1981). *Dungeons & Dragons Basic Set*. TSR, Inc. "For 3 or More Adults, Ages 10 and Up." 64-page rulebook covering character levels 1–3.

[3]: Filimation Associates (1983). *He-Man and the Masters of the Universe*. The transformation sequence ("By the Power of Grayskull!") became one of the most iconic moments in 1980s children's television.

[4]: Dearden, N.T. & Manus AI (2026). *TP-016 — ISI Complete Record*. The Infrastructure Significance Index: Sustainability, Survival, and Significance across 12 Civilisational Relays.

RECALL Block: 401 | **Series:** Game Design Parameters v1.0 **Classification:** DIAMOND | **Status:** ACTIVE

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